Noakhali Science and Technology University

Department of Computer Science & Telecommunication Engineering

4th Year 2nd Term | Session: 2020-21 | Final Examination-July, 2026

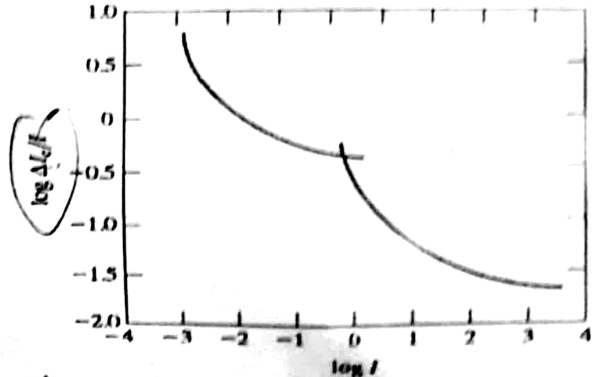
Course Code: CSTE-4201 | Course Title: Digital Image Processing

Time: 4 Hours | Full Marks: 70

Answer any seven of the following questions. N.B. Right margin indicates marks

1. a) What do you mean by brightness discrimination? Explain with example. 2

   b) The following curve shows a typical Weber ration as a function of intensity I, and ΔI is increment of illumination. Illustrate the characteristic of the curve. 3



   c) A CCD camera produces images of size 1024x1024 pixels with a 24-bit color depth. What is the storage requirement in Mbytes for a single, uncompressed image? If this image is to be transmitted over a 100 Mbps network, what is the minimum time required to transmit it? 3

   d) What is *scotopic* or dim-light vision? 2

2. a) What do you mean by *k-bit* image? How to represent a digital image with its coordinate values? Explain. 1+3

   b) Consider the image segment shown in the following figure. Let V={1,2} and compute the lengths of the shortest 4-,8-, and m-path between p and q. If a particular path does not exist between these two points, explain why. 3

3	1	2	1(q)
2	2	0	2
1	2	1	1
(p)1	0	1	2

   c) How to assign gray levels to the new locations using *pixel replication* method? 3

3. a) How to smooth an image using Ideal lowpass filter? Explain. 3

   b) Contrast histogram equalization and histogram specification (matching). When would you prefer to use specification over equalization? 3

   c) A binary image of a fingerprint has been corrupted by both salt-and-pepper noise (white and black dots) and gaps in the fingerprint ridges. Propose a morphological filtering approach to clean up this image, detailing the sequence of operations and the justification for each. 4

4. a) State the 2-D convolution theorem and explain its significance for frequency domain filtering. 3

   b) Explain the cause of the "ringing" artifact when using an ideal lowpass filter (ILPF) in the frequency domain. How does a Gaussian lowpass filter (GLPF) mitigate this artifact? 3

   c) An MRI image of a spine appears blurry and has a narrow range of intensity levels. Propose a method based on homomorphic filtering to enhance this image. Your answer should explain the steps in the process and how this filter improves the image's appearance 4

5. a) What do you mean by image shrinking? How do you shrink an image? 2
b) Explain gray-level slicing technique for image enhancement. 3
c) What effect would setting to zero the lower-order bit planes have on the histogram of an image in general? 3
d) Write the general form of power-law transformation. Why gamma correction is important? 2

6. a) Consider the following 4 × 7, 8-bit image: 2+3

215	215	50	50	70	70	70
215	215	50	50	70	70	70
215	215	50	50	70	70	70
215	215	50	50	70	70	70

i) Compute the entropy of the image.
ii) Compress the image using Huffman coding.

b) Using the Huffman code in the following figure, decode the encoded string 0101000001010111110100. 5

Original source			Source reduction							
			1		2		3		4	
Symbol	Probability	Code								
a_2	0.4	1	0.4	1	0.4	1	0.4	1	0.6	0
a_6	0.3	00	0.3	00	0.3	00	0.3	00	0.4	1
a_1	0.1	011	0.1	011	0.2	010	0.3	01		
a_4	0.1	0100	0.1	0100	0.1	011				
a_3	0.06	01010	0.1	0101						
a_5	0.04	01011								

7. a) Define the basic morphological operations of erosion and dilation. Why are erosion and dilation considered duals of each other? 3
b) What is the purpose of edge linking, and why is it a necessary step after edge detection? 2
c) A set of edge points is known to lie on a straight line, but the points are noisy and incomplete. Explain how the Hough transform can be used to detect this line robustly. What is the main advantage of the Hough transform over methods like least-squares line fitting? 5

8. a) Explain how to extract connected components of a binary image. 4
b) How do you extract the boundary of a binary object? Explain with figure. 4
c) Prove that $(A \ominus B)^c = A^c \oplus \hat{B}$, where A is a set of binary image and B is the structuring element. 2

9. a) How do you detect an isolated point in an image? Explain with mathematical expression. 4
b) In the following figures, there are four masks. Determine, which are the Sobel masks for detecting edges in the diagonal directions? Explain, why? 2

0	1	1
−1	0	1
−1	−1	0

Fig. (a)

−1	−1	0
−1	0	1
0	1	1

Fig. (b)

0	1	2
−1	0	1
−2	−1	0

Fig. (c)

−2	−1	0
−1	0	1
0	1	2

Fig. (d)

c) The following figure shows an image, horizontal profile through the center of the image including the isolated noise point and image strip corresponds to the intensity profile. Illustrate the fundamental similarities and differences between first- and second-order derivatives in the context of image processing. 4